Global MAP-Optimality by Shrinking the Combinatorial Search Area with Convex Relaxation

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Heidelberg Collaboratory for Image Processing

MRF Energy Minimization

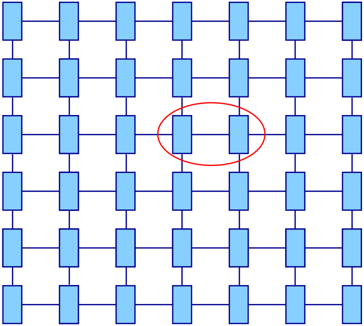
$$\min_{x\in\mathcal{X}} E(x) := \min_{x\in\mathcal{X}} \sum_{v\in\mathcal{V}} \theta_v(x_v) + \sum_{uv\in\mathcal{E}} \theta_{uv}(x_u, x_v)$$

- **Segmentation** [Rother et al. 2004], [Nowozin, Lampert 2010]
- **Multi-camera stereo** [Kolmogorov, Zabih 2002]
- **Stereo and Motion** [Kim et al. 2003]
- **Clustering** [Zabih, Kolmogorov. 2004]
- **Medical imaging** [Raj et al. 2007]
- **Pose Estimation** [Bergtholdt et al. 2010], [Bray et al. 2006]
- ...

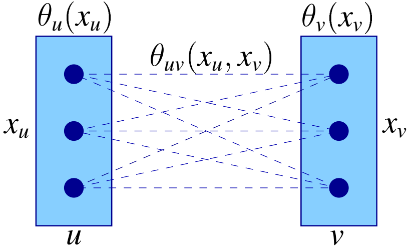
Computer Vision energy minimization benchmarks:
[Szeliski et al. 2008], [Kappes et al. CVPR, 2013]

MRF Energy Minimization

$$\min_{x \in \mathcal{X}} E(x) := \min_{x \in \mathcal{X}} \sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$



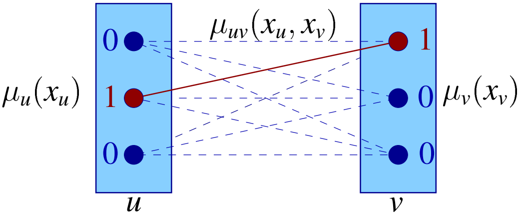
graph $(\mathcal{V}, \mathcal{E})$



Integer LP Formulation

$$\min_{\mu \geqslant 0} \sum_{v \in \mathcal{V}} \sum_{x_v \in \mathcal{X}_v} \theta_v(x_v)\mu_v(x_v) + \sum_{uv \in \mathcal{E}} \sum_{x_u, x_v \in \mathcal{X}_{uv}} \theta_{uv}(x_u, x_v)\mu_{uv}(x_u, x_v)$$

$$\text{s.t.} \quad \begin{aligned} & \textstyle\sum_{x_v \in \mathcal{V}} \mu_v(x_v) = 1, \ v \in \mathcal{V} \\ & \textstyle\sum_{x_v \in \mathcal{V}} \mu_{uv}(x_u, x_v) = \mu_u(x_u), \ x_u \in \mathcal{X}_u, \ uv \in \mathcal{E} \\ & \textstyle\sum_{x_u \in \mathcal{V}} \mu_{uv}(x_u, x_v) = \mu_v(x_v), \ x_v \in \mathcal{X}_v, \ uv \in \mathcal{E}\,. \\ & \mu \in \{0, 1\}^N \end{aligned}$$



~~Integer~~ LP Formulation

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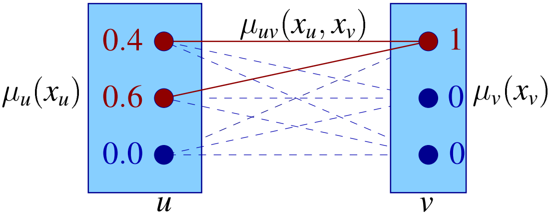
$$\text{s.t.} \quad \begin{array}{l} \sum_{x_v \in \mathcal{V}} \mu_v(x_v) = 1, \ v \in \mathcal{V} \\ \sum_{x_v \in \mathcal{V}} \mu_{uv}(x_u, x_v) = \mu_u(x_u), \ x_u \in \mathcal{X}_u, \ uv \in \mathcal{E} \\ \sum_{x_u \in \mathcal{V}} \mu_{uv}(x_u, x_v) = \mu_v(x_v), \ x_v \in \mathcal{X}_v, \ uv \in \mathcal{E}\,. \\ \cancel{\mu \in \{0,1\}^N} \ \mu \in [0,1]^N \end{array}$$



LP Relaxation: typical solution

color segmentation problem

integer and fractional labelings

- Is the integer part of the solution correct?
- In general - NO! In practice - YES.
- Proof?

Related Approach: Partial Optimality

QPBO:[Hammer et al. 1984],[Boros, Hammer 2002],[Rother et al. 2007], [Kohli, Shekhovtsov et al. 2008],[Windheuser et al. 2012], [Kahl,Strandmark 2012];
Submodular relaxation:[Kovtun 2003], [Kovtun PhD Thesis 2005], [Shekhovtsov,Hlavač 2011];
LP relaxation:[Swoboda, Savchynskyy et al. 2013].

integer and fractional labeling | solved | partial solution

Our approach | [Kovtun 2003]

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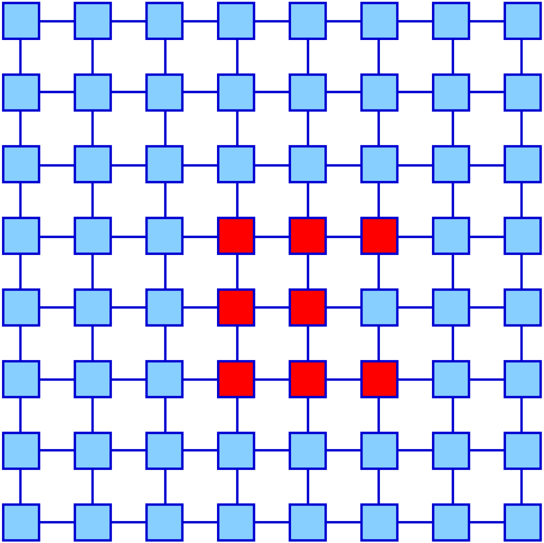
integer and fractional labeling $\Rightarrow$ NOT solved | partial solution

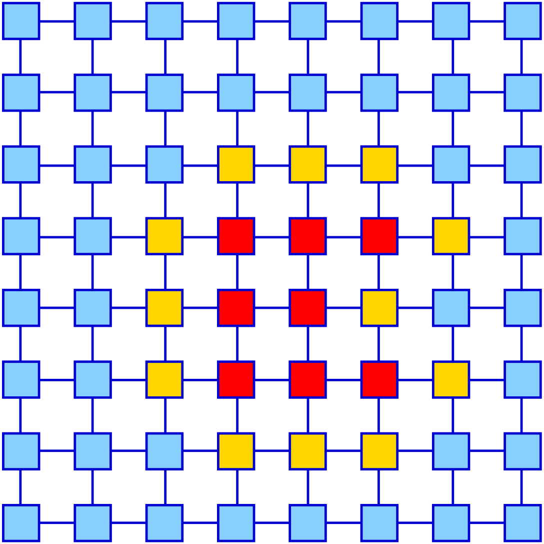
Our approach | [Kovtun 2003]

Main Idea



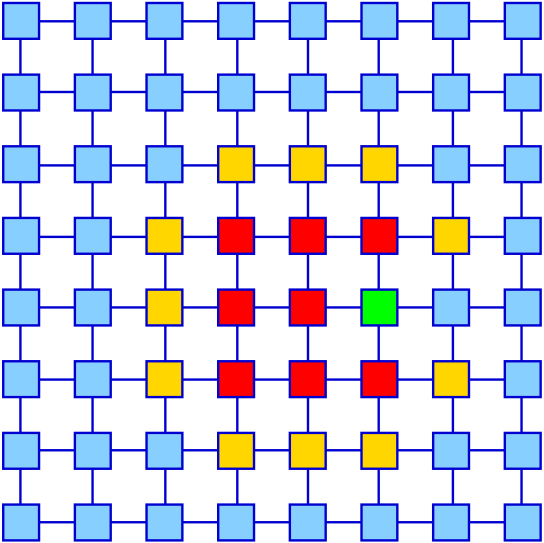
blue = LP-tight, red = ILP (combinatorial)

Main Idea



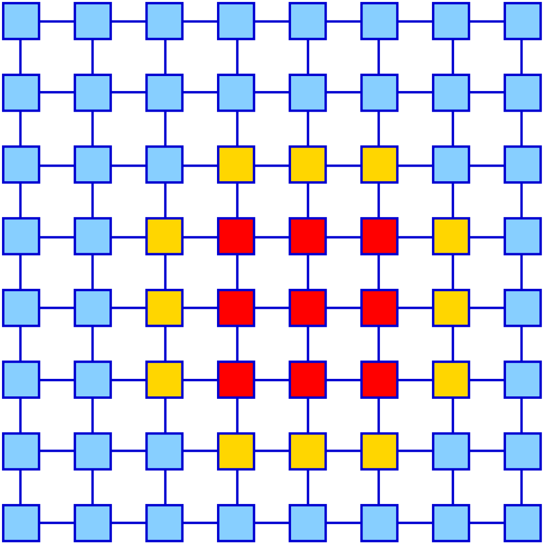
Add border and solve 'blue' with LP and 'red+yellow' with ILP

Main Idea



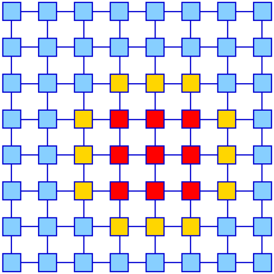
Check whether LP and ILP labelings are consistent on the border

Main Idea



Increase ILP (combinatorial) set and repeat

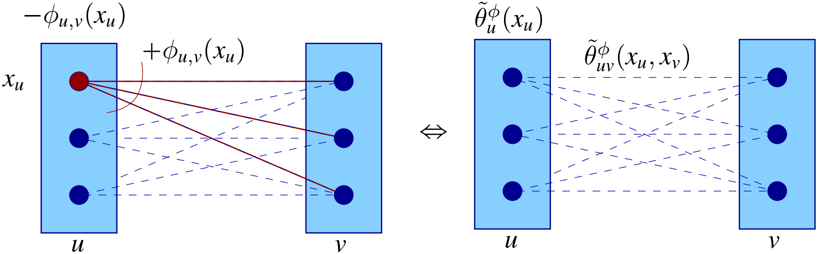
From Idea to Algorithm



- Is the consistency on the border sufficient for optimality?
- How to select the initial LP/ILP splitting?
- How to encourage consistency on the border?
- How to avoid re-solving LP-tight part?
- Do we need to solve LP-tight part to optimality?

Background: Reparametrization (Equivalent transformations)

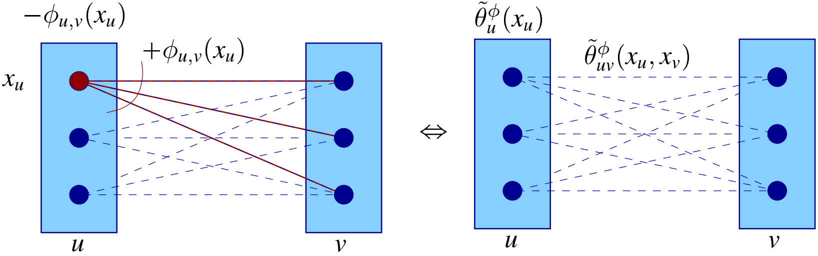
$$\sum_{v\in\mathcal{V}} \theta_v(x_v) + \sum_{uv\in\mathcal{E}} \theta_{uv}(x_u, x_v) \equiv \sum_{v\in\mathcal{V}} \tilde{\theta}_v^{\phi}(x_v) + \sum_{uv\in\mathcal{E}} \tilde{\theta}_{uv}^{\phi}(x_u, x_v)$$



Background: Reparametrization, Dual problem

$$\text{Primal:} \quad E(x) = \min_x \sum_{v\in\mathcal{V}} \theta_v(x_v) + \sum_{uv\in\mathcal{E}} \theta_{uv}(x_u, x_v)$$

$$\text{Dual:} \quad D(\phi) = \max_\phi \sum_{v\in\mathcal{V}} \min_{x_v} \tilde{\theta}_v^\phi(x_v) + \sum_{uv\in\mathcal{E}} \min_{x_{uv}} \tilde{\theta}_{uv}^\phi(x_{uv})$$



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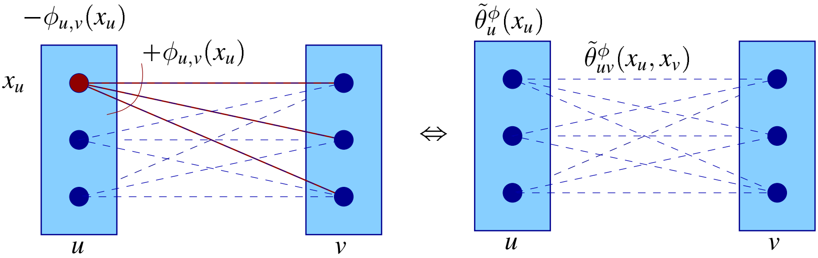
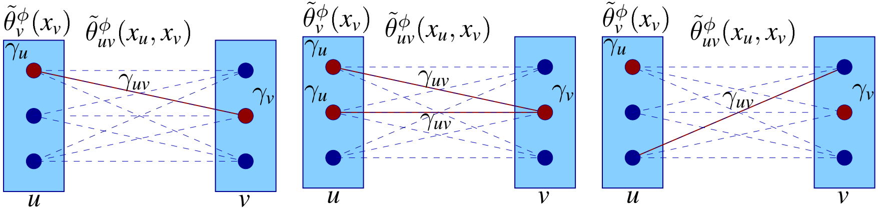
$$D(\phi) \leqslant E(x)$$



Background: Arc Consistency

Dual: $$D(\phi) = \max_{\phi} \sum_{v \in \mathcal{V}} \underbrace{\min_{x_v} \tilde{\theta}_v^{\phi}(x_v)}_{\gamma_v} + \sum_{uv \in \mathcal{E}} \underbrace{\min_{x_{uv}} \tilde{\theta}_{uv}^{\phi}(x_{uv})}_{\gamma_{uv}}$$



strict arc consistency | ~~strict~~ arc consistency | ~~arc consistency~~

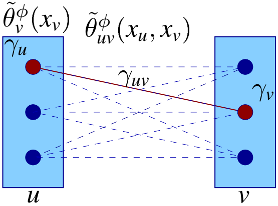
Background: Trivial Problem, Strict Arc Consistency

Theorem.

$$\text{Strict arc consistency in all nodes} \\ \Downarrow \\ \text{the non-relaxed problem is solved.}$$

Proof. Strict arc consistency $\Rightarrow D(\phi) = E(x^*)$, x^* consists of $\gamma_v,\ \gamma_{uv}$.

$D(\phi) \leqslant E(x) \Rightarrow x^*$ is the solution.



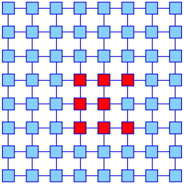
strict arc consistency

LP Relaxation: typical (approximate) solution

Blue - strictly arc consistent, red - otherwise.

Algorithm

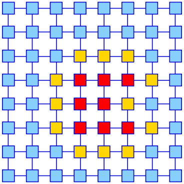
0) Initialize:



Solve LP relaxation and reparametrize: $\theta \rightarrow \tilde{\theta}^{\phi}$
'Blue' = the strictly arc consistent nodes.

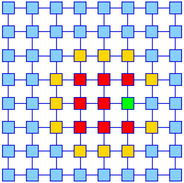
t) Iterate until consistent solution on the border:

→



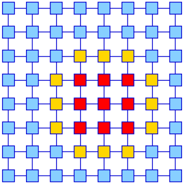
apply combinatorial solver

→



check border consistency

→

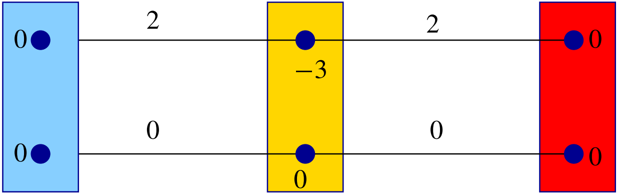


increase combinatorial set if needed

→

Why reparametrize?

Theorem. If the energy minimization of 'blue+yellow' is trivial, border consistency is sufficient for optimality.



!NB

Triviality is actually required (counterexample due to A. Shekhovtsov)

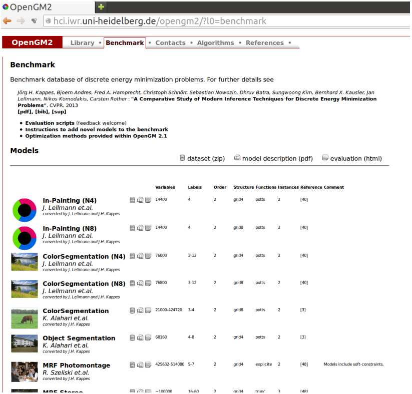
Why reparametrize?

- Optimality condition
- Initial splitting criterion
- Encouraging of boundary consistency
- Potentially speeds-up combinatorial solvers (LP presolve)

Important, that

- An LP solver needs to be executed only *once*
  Because due to strict consistency local decisions are optimal: nodes removal does not change the remaining part of the solution
- *Suboptimal* reparametrization can be used as well
  Because we did not use optimality of the reparametrization

Experimental Evaluation: OpenGM Library and Benchmark


OpenGM2

hci.iwr.uni-heidelberg.de/opengm2/?l0=benchmark

OpenGM2 | Library • Benchmark • Contacts • Algorithms • References •

Benchmark

Benchmark database of discrete energy minimization problems. For further details see

Jörg H. Kappes, Bjoern Andres, Fred A. Hamprecht, Christoph Schnörr, Sebastian Nowozin, Dhruv Batra, Sungwoong Kim, Bernhard X. Kausler, Jan Lellmann, Nikos Komodakis, Carsten Rother : **"A Comparative Study of Modern Inference Techniques for Discrete Energy Minimization Problems"**, CVPR, 2013
[pdf], [bib], [sup]

- **Evaluation scripts** (feedback welcome)
- **Instructions to add novel models to the benchmark**
- **Optimization methods provided within OpenGM 2.1**

Models

dataset (zip) model description (pdf) evaluation (html)

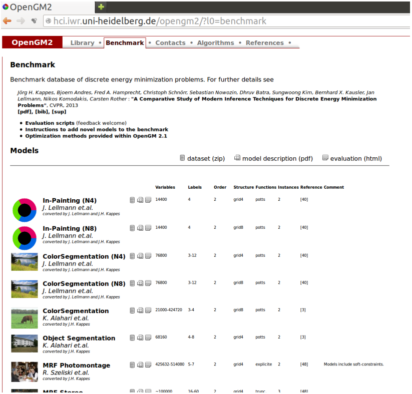
	Variables	Labels	Order	Structure	Functions	Instances	Reference	Comment
In-Painting (N4) *J. Lellmann et.al.* *converted by J. Lellmann and J.H. Kappes*	14400	4	2	grid4	potts	2	[40]	
In-Painting (N8) *J. Lellmann et.al.* *converted by J. Lellmann and J.H. Kappes*	14400	4	2	grid8	potts	2	[40]	
ColorSegmentation (N4) *J. Lellmann et.al.* *converted by J. Lellmann and J.H. Kappes*	76800	3-12	2	grid4	potts	2	[40]	
ColorSegmentation (N8) *J. Lellmann et.al.* *converted by J. Lellmann and J.H. Kappes*	76800	3-12	2	grid8	potts	2	[40]	
ColorSegmentation *K. Alahari et.al.* *converted by J.H. Kappes*	21000-424720	3-4	2	grid8	potts	2	[3]	
Object Segmentation *K. Alahari et.al.* *converted by J.H. Kappes*	68160	4-8	2	grid4	potts	2	[3]	
MRF Photomontage *R. Szeliski et.al.* *converted by J.H. Kappes*	425632-514080	5-7	2	grid4	explicite	2	[48]	Models include soft-constraints.
MRF Stereo	~100000	16-60	2	grid4	trunc.	3	[48]	

Open Library for Graphical Models:

- inference algorithms;
- benchmark data;
- includes Middlebury MRF benchmark
- comparison tables;

Just type `'OpenGM'` in Google...

Experimental Evaluation: OpenGM Library and Benchmark



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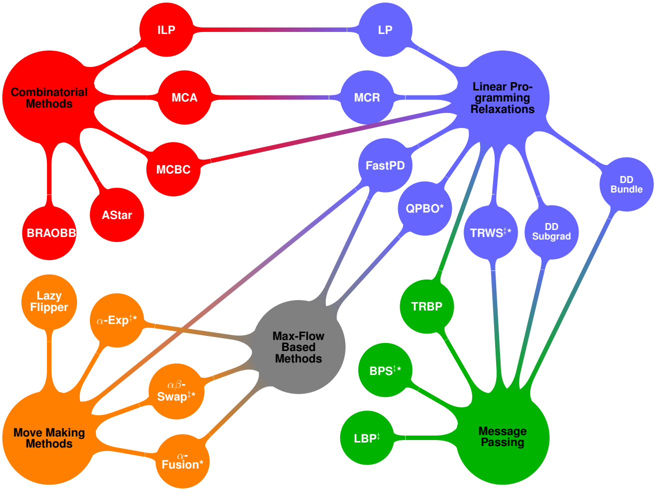
Just type `'OpenGM'` in Google...

Our code will be freely available as a part of OpenGM soon!

Experimental Evaluation: Used Methods

Algorithms, available in OpenGM:



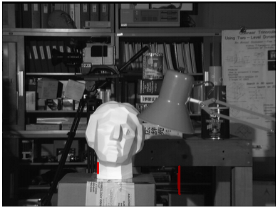
We used:

- **TRW-S** [Kolmogorov 2005] as LP solver
- **CPLEX** [IBM] as ILP solver.

Middlebury MRF Benchmark

tsukuba

384×288, 16 labels



$$E_{min} = 369218$$
$$E_{TRWS} = 369218$$

venus

434×383, 20 labels

$$E_{min} = 3048043$$
$$E_{TRWS} = 3048296$$

family

752×566, 5 labels

$$E_{min} = 184813$$
$$E_{TRWS} = 184825$$

Middlebury MRF Benchmark

teddy

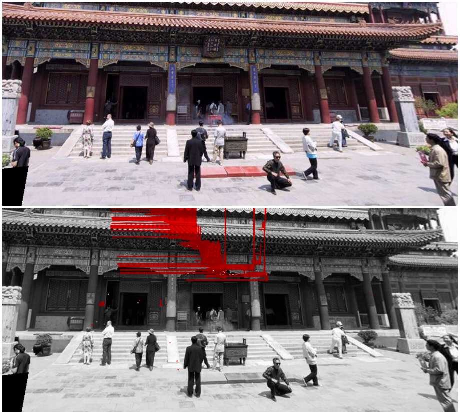
450×375, 60 labels

1 iteration of ILP

= out of memory

panorama

1071×480, 7 labels



1 iteration of ILP

= out of memory

Color Segmentation: 26 Potts models

Solved

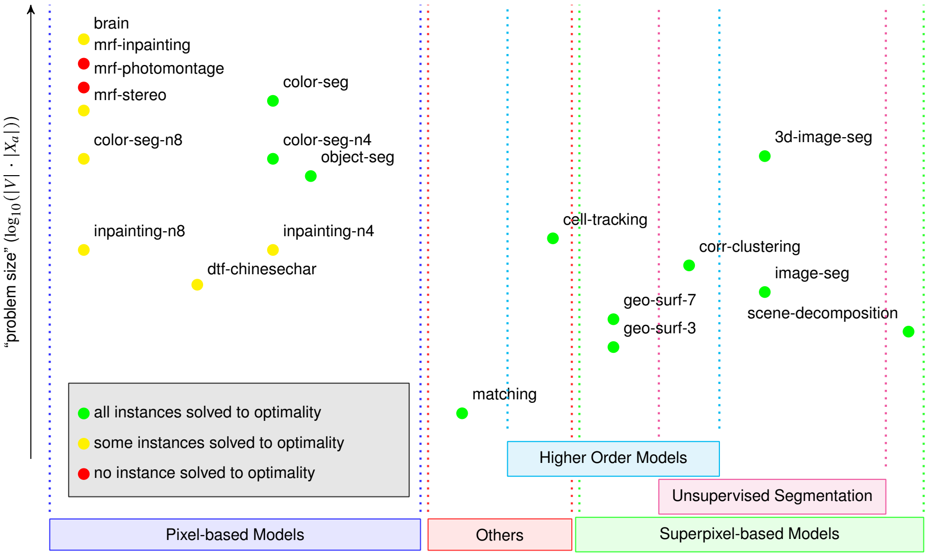
Potts Models: Comparison to State-of-the-Art

Dataset	Our, sec	Multiway cut, sec	TRWS/CPLEX, sec
pfau-n4	293	10584	281/12
pfau-n8	545	> 55496	523/17
palm-n4	51	197	32/9
palm-n8	133	569	59/74

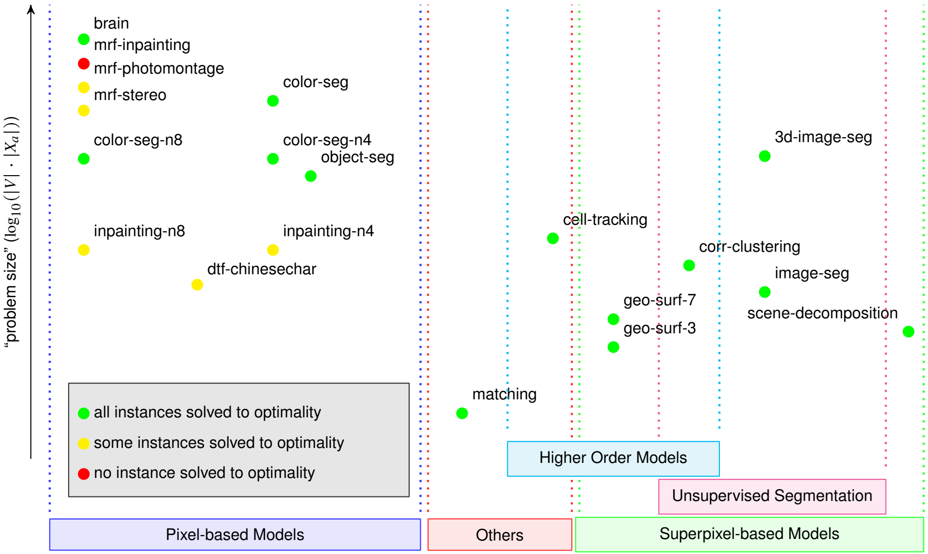
Table: Processing time

Multiway cut: [Kappes et al. 2011],[Kappes et al. 2013]

OpenGM Models: w/o our results



OpenGM Models: with our results



Conclusions and Future Work

Our approach

- does efficient extraction of the complex, combinatorial subproblem;
- is generic: allows almost any combination of LP and ILP solvers;
- makes the problems, which are easy in practice, easy in theory.

Limitations:

- sparse graphs;
- LP relaxation is *almost* tight.

Future work:

- Alternative and specialized solvers for LP and ILP.
- Higher order models.
- Tighter convex relaxations.

Proof of the Main Theorem

Definition $\mathcal{V}_\mathcal{A} \subset \mathcal{V}$, $\mathcal{V}_\mathcal{C} = \{v \in \mathcal{V}_\mathcal{A} : \exists uv \in \mathcal{E}_\mathcal{G} : u \in \mathcal{V}_\mathcal{G} \backslash \mathcal{V}_\mathcal{A}\}$, $\mathcal{V}_\mathcal{B} = \mathcal{V}_\mathcal{C} \cup (\mathcal{V}_\mathcal{G} \backslash \mathcal{V}_\mathcal{A})$, $\mathcal{Q} = (\mathcal{V}_\mathcal{Q}, \mathcal{E}_\mathcal{Q})$, $\mathcal{E}_\mathcal{Q} = \{uv \in \mathcal{E}_\mathcal{G} : u, v \in \mathcal{V}_\mathcal{Q}\}$.

Theorem. Let

- $x^*_\mathcal{A}$ and $x^*_\mathcal{B}$ minimize the energy on $\mathcal{A}$ and $\mathcal{B}$ resp.,
- $x^*_\mathcal{A}|_\mathcal{C} = x^*_\mathcal{B}|_\mathcal{C}$,
- problem $\min_{x_\mathcal{A}} E_\mathcal{A}(x_\mathcal{A})$ is trivial.

Then $x^* = (x^*_\mathcal{A}, x^*_\mathcal{B}|_{\mathcal{B}\backslash\mathcal{C}})$ is optimal on $\mathcal{G}$.

Proof. $E_\mathcal{G}(x) \to E^\theta_\mathcal{G}(x)$

$$\theta'_w(x_w) := \begin{cases} 0, & w \in \mathcal{V}_\mathcal{C} \cup \mathcal{E}_\mathcal{C} \\ \theta_w(x_w), & w \notin \mathcal{V}_\mathcal{C} \cup \mathcal{E}_\mathcal{C} \end{cases}$$

$E^\theta_\mathcal{G}(x) = E^{\theta'}_\mathcal{A}(x_\mathcal{A}) + E^\theta_\mathcal{B}(x_\mathcal{B})$

$\min_{x_\mathcal{A}} E_\mathcal{A}(x_\mathcal{A})$ is trivial $\Rightarrow x^*_\mathcal{A} \in \arg\min_{x_\mathcal{A}} E^{\theta'}_\mathcal{A}(x)$

$$\min_x E^\theta_\mathcal{G}(x) = \{\min_{x_\mathcal{A}, x_\mathcal{B}} E^{\theta'}_\mathcal{A}(x_\mathcal{A}) + E^\theta_\mathcal{B}(x_\mathcal{B}) |\ \text{s.t. } x_\mathcal{A}|_\mathcal{C} = x_\mathcal{B}|_\mathcal{C}\}$$

$$= \min_{x'_\mathcal{C}} \min_{x_\mathcal{A}\,:\, x_\mathcal{A}|_\mathcal{C} = x'_\mathcal{C}} E^{\theta'}_\mathcal{A}(x_\mathcal{A}) + \min_{x_\mathcal{B}\,:\, x_\mathcal{B}|_\mathcal{C} = x'_\mathcal{C}} E^\theta_\mathcal{B}(x_\mathcal{B})$$

$$\geqslant \min_{x_\mathcal{A}} E^{\theta'}_\mathcal{A}(x_\mathcal{A}) + \min_{x_\mathcal{B}} E^\theta_\mathcal{B}(x_\mathcal{B}) = E^{\theta'}_\mathcal{A}(x^*_\mathcal{A}) + E^\theta_\mathcal{B}(x^*_\mathcal{B}) = E^\theta_\mathcal{G}(x^*)$$